

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (Biotechnology) (Credit System) Degree Examinations**

May - June 2023

20BT602 - UPSTREAM PROCESSING TECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **Unit – I** and **Two full** questions from **Unit – II & Unit – III** each.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the isolation and screening of animal cells.	10	L*1	1	2
	b) Describe the various methods to improve microbial cells.	10	L2	1	2
2.	a) Explain the improvement of plant cell by somatic hybridization.	10	L2	1	2
	b) Explain the preservation method for cell and tissues.	10	L3	1	2
Unit – II					
3.	a) Explain media used for microbial cell culture.	10	L2	2	2
	b) Explain methods of asepsis employed in plant cell culture.	10	L3	3	2
4.	a) Discuss sterilization of media by continuous process.	10	L3	3	2
	b) Explain media optimization with Placket Burman design and Factorial design.	10	L2	3	2
5.	a) Explain the design of batch sterilization.	10	L4	3	2
	b) Write note on (i) oxygen requirement (ii) metabolic regulator in media.	10	L2	2	2
Unit – III					
6.	a) Explain the growth curve in plant and bacterial cells.	10	L2	5	2
	b) Explain the method of inoculum development for fungal culture.	10	L2	4	2
7.	a) Discuss on plant cell suspension culture.	10	L2	4	2
	b) Write note on (i) yeast inoculum preparation (ii) criteria for inoculum transfer.	10	L2	5	2
8.	a) Discuss on the Monod kinetic model for Substrate and Biomass.	10	L2	5	2
	b) Write a note on hairy root culture.	10	L2	4	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
